

Popularity Ratio Maximization: Surpassing Competitors through Influence Propagation

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ABSTRACT

In this paper, we present an algorithmic study on how to surpass competitors in popularity by strategic promotions in social networks. We first propose a novel model PA-IC, in which we integrate the Preferential Attachment (PA) model for popularity growth with the Independent Cascade (IC) model for influence propagation in social networks. In PA-IC, a popular item and a novice item grab shares of popularity from the natural popularity growth via the PA model, while the novice item tries to gain extra popularity via influence cascade in a social network. The *popularity ratio* is defined as the ratio of the popularity measure between the novice item and the popular item. We formulate *Popularity Ratio Maximization (PRM)* as the problem of selecting seeds in multiple rounds to maximize the popularity ratio in the end. We analyze the popularity ratio and show that it is monotone but not submodular. To provide an effective solution, we design a surrogate objective function and show that empirically it is very close to the original objective function while theoretically, it is monotone and submodular. We design two efficient algorithms, one for the overlapping influence and non-overlapping seeds (across rounds) setting and the other for the non-overlapping influence and overlapping seed setting, and further discuss how to deal with other model and problem variants. Our empirical evaluation further demonstrates that the proposed PRM-IMM method consistently achieves the best popularity promotion compared to other methods. Our theoretical and empirical analyses shed light on the interplay between influence maximization and preferential attachment in social networks.

CCS CONCEPTS

• Information systems → Data mining; Computational advertising; Social networking sites.

KEYWORDS

graph algorithms, preferential attachment, popularity ratio maximization

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1 INTRODUCTION

Influence maximization (IM) is the problem of finding a subset of nodes (seed nodes) in a social network that could maximize the spread of influence [17]. It is a well-studied problem with applications on viral marketing, information propagation monitoring and control, etc (cf. [8, 22]). While most studies treat influence maximization as a stand-alone problem for viral marketing, in this paper, we want to explore the means of using influence maximization to boost the popularity of an item (an idea, a product, etc) and surpass a competitor in a natural growth environment.

Consider the following hypothetical motivating example. Alice opened her new restaurant, Café Alice, at a local shopping mall. However, a popular restaurant Bob's Kitchen is in the same mall. Without promotion, Alice would definitely lose many customers to Bob's Kitchen. Thus Alice wants to distribute free-dish coupons to selected people, hoping them to propagate the information about Café Alice and attract more people to her restaurant, boosting the popularity of her restaurant in a short period of time to catch up or even surpass Bob's Kitchen. Similar situations occur in online social media, such as a new blogger or podcaster trying to increase popularity through promotion in social networks. Different from pure viral marketing, popularity growth has a natural rich-get-richer effect independent of the social network, meaning that the popular item would naturally attract more customers or users. For example, customers typically check nearby restaurants' popularity ratings on mobile apps such as Yelp for restaurant selection, making more people visit the popular restaurant. This effect certainly gives a hard time for the new restaurant to catch up. Thus, incentive promotion by the new restaurant needs to incorporate this factor when deciding on viral marketing strategies.

This paper integrates the natural popularity growth with the incentive promotion into one coherent model to solve the above problem. In particular, the preferential attachment (PA) [1, 27, 35] is a well-known model for the natural rich-get-richer effect of popularity growth, while the independent cascade (IC) model [17] is a classical model for influence propagation. We integrate these two models into the novel PA-IC model to characterize the combined effect of natural popularity growth and incentive promotion. More specifically, in a multi-round setting, we model the popularity growth of two items: one novice item and one popular item. We use d_t^p and d_t^n to denote the popularity of the popular and novice items respectively, at the end of round t . In each round, a number of random customers pick from these two items with probability proportional to their popularity, realizing the preferential attachment model. Meanwhile, the novice item executes a promotion plan, selecting a set of seeds S_t in round t for T consecutive rounds, and the influence spread generated by the seed set S_t would add to the popularity measure of the novice item at the end of round t . We define $r_t = d_t^n / d_t^p$ as the *popularity ratio* in round t . The promotion task of the novice item is defined as the following *popularity ratio*

maximization (PRM) problem: Given (a) the social network and its IC model parameters, (b) the initial popularity measures d_0^p and d_0^n , and (c) a promotion budget k , find k seeds and allocate them into T consecutive rounds such that the popularity ratio r_T at the end of round T is maximized. Depending on whether we allow overlapping seeds (OS) across rounds and whether repeated activation of the same nodes in different rounds are repeatedly counted (overlapping influence, OI), we further consider different settings such as OINS and NIOS.

For both OINS and NIOS settings, we derive the formula for the popularity ratio and show that it is monotone but not submodular, indicating that direct optimization on this objective function may be complex. To provide an effective solution, we simplify the objective function and obtain a surrogate function and show that empirically it is very close to the original objective function while theoretically, it is monotone and submodular. We then design two efficient algorithms based on the reverse influence sampling approach [3] and the IMM algorithm [31] to solve this surrogate problem for the two settings, and prove their theoretical guarantees. We further discuss how to handle other model and problem variants, such as when the popular item also conducts promotion, and when the novice item want to use the minimum budget to surpass the popular item in a given time limit. Our empirical evaluation demonstrates that the proposed PRM-IMM method always achieves the best popularity promotion compared to other methods.

One important point we demonstrate both analytically and empirically is that the seed allocation of the novice item is tightly dependent on the natural growth of the environment as well as the popularity of the popular item. This point indicates that the PRM task is quite different from the classical influence maximization — the promotion of the novice item has to consider the popularity growth and its relative position against the popular item, while the classical influence maximization only targets at increasing one's own influence spread or popularity.

In summary, our contributions include: (a) proposing the PA-IC model that integrates preferential attachment with independent cascade propagation model; (b) defining the PRM problem and studying its properties; and (c) designing an efficient algorithm to solve the PRM problem. To the best of our knowledge, this is the first study that integrates preferential attachment and an influence propagation model into a coherent popularity growth model and provides an algorithmic study on a class of optimization problems under such a model.

2 RELATED WORKS

Preferential Attachment. Yule firstly considered using the preferential attachment to explain the power-law distribution of flowering plants [35]. A clearer and more general development of how preferential attachment leads to a power law was given by Simon [27]. The Barabási-Albert (BA) model shows that the power-law degree distribution in the real networks can be produced by the combination of growth and preferential attachment [1]. Preferential attachment is now the standard model for the rich-get-richer effect.

Influence Maximization. Domingos and Richardson first study influence maximization (IM) [12, 26]. Then it is formulated as an optimization problem by Kempe et al. [17], who define the famous

independent cascade and linear threshold models, show that influence maximization under these models are NP-hard, and connect the problem to monotone and submodular function maximization. Influence maximization has been extensively studied (see surveys [8, 22]). One direction is to improve the scalability of the algorithms [3, 9–11, 13, 16, 18, 20, 34]. The state-of-the-art scalable solution is based on the reverse influence sampling (RIS) approach first proposed by Borg et al. [3], and later improved and refined by a series of studies [6, 25, 30–32]. Our algorithms are also based on the RIS approach and is adapted from the IMM algorithm of [31].

Another related direction is competitive influence maximization for multiple items [4, 8, 14, 15, 21, 23]. However, all these studies focus on the competitive diffusion of multiple items in social networks. In contrast, the competition in our model stems from the popularity growth dictated by preferential attachment, and influence maximization is only a tool for the novice item to increase popularity. To the best of our knowledge, we are the first to consider popular growth via both the PA effect and network diffusion, and study their nontrivial interplay in an optimization problem.

3 MODEL AND PROBLEM DEFINITION

In this section, we introduce the PA-IC model, which integrates the preferential attachment (PA) mechanism for popularity growth with the independent cascade (IC) model for influence propagation. The PA mechanism models show two items, a popular item and a novice item, divide the shares of popularity from the natural growth of the customer base, while the novice item further utilizes the IC model propagation to promote its popularity in order to catch up with the popular item. Based on the PA-IC model, we define the optimization problem of *popularity ratio maximization*, which characterizes how to allocate the promotional budget of the novice item to achieve the best promotion results.

3.1 PA-IC Model

Preferential attachment (PA) [1] is a well-known model to characterize the rich-get-richer effect in the growth of networks or popularity. The basic form of the PA model is as follows. Let d_t^i be the popularity measure of item i at time t . When a number of a users come at time t , each user selects item i with probability proportional to d_t^i . Thus popularity grows proportionally, i.e., $d_{t+1}^i = d_t^i + a \cdot d_t^i / \sum_j d_t^j$, where a is the natural growth parameter in PA-IC model.

Beyond the natural growth of popularity governed by the PA mechanism, we would like to incorporate the influence propagation model for popularity promotion. Independent cascade (IC) model [17] is a classical diffusion model widely adopted in the information and influence propagation literature. In the IC model, a social network is modeled as a directed graph $G = (V, E, p)$, where V is the set of $N = |V|$ nodes representing individuals, E is the set of $M = |E|$ directed edges representing influence relationships between pairs of individuals, and $p : E \rightarrow (0, 1]$ gives the influence probability on every edge, i.e., $p(u, v)$ is the influence probability on edge $(u, v) \in E$. Nodes have two states, active and inactive. Nodes that have been activated will always remain active. The propagation process starts from a seed set $S \subset V$ in discrete time steps. At step $\tau = 0$, only the nodes in S are active. At step $\tau \geq 1$, a node u

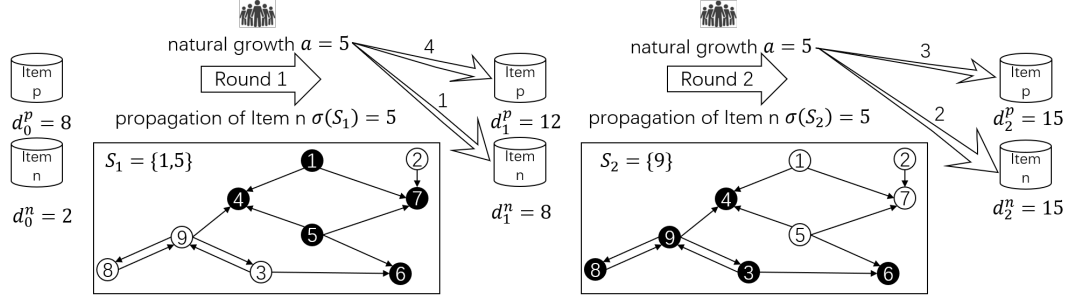


Figure 1: A numerical example of the PA-IC model with the overlapping influence setting. In this example, we show the changes in popularity measure of two items, within two rounds. In this directed graph, every edge's activation probability is equal to 1. Black nodes is the "active" nodes in each round.

activated at step $\tau - 1$ attempts to activate each of its inactive out-neighbor v with success probability $p(u, v)$. Propagation terminates when no more nodes are activated. A key measure is the *influence spread* of the seed set S , denoted as $\sigma(S)$, which is the expected number of activated nodes at the end of propagation starting from the seed set S . The IC model has the following equivalent live-edge graph description. A random *live-edge graph* L is sampled from $G = (V, E, p)$ such that every edge $(u, v) \in E$ has an independent probability of $p(u, v)$ to be included in L . Then given a seed set S , the set of nodes activated in a stochastic diffusion from S is the set of nodes reachable from S in the live-edge graph L , denoted as $\Gamma(S, L)$. Therefore, we have $\sigma(S) = \mathbb{E}_L[|\Gamma(S, L)|]$.

We integrate the PA model with the IC model in the following PA-IC model to characterize both the natural growth and the promotional growth of the popularity. For simplicity, this paper focuses on two competing items: a popular item and a novice item, since all items grow proportionally in the PA model. The popular item starts with a higher popularity measure, while the novice item starts with a lower one. If only natural growth is available, the novice item would never catch up with the popular item, and the gap would only be widened due to the rich-get-richer effect of the PA model. To catch up and surpass the popular item, the novice item needs to use incentive promotion in the social network to increase its popularity. The technical description of the PA-IC model to cover the above aspects is given below.

In the PA-IC model, we model T rounds of promotion and influence propagation. Let d_t^p and d_t^n denote the popularity measure of the popular and novice items respectively, at the end of round t . In each round $t \in [T]$, some customers would naturally select between the novice or popular item. We use natural growth parameter a to denote the number of customers. These a customers select between novice and popular items according to the PA model.

Besides the natural growth, in each round t , the novice item would select a seed set S_t for its promotion, and these seeds will propagate influence throughout the social network G following the IC model. The expected number of influenced nodes $\sigma^\square(S_t)$ is also added to the popularity measure of the novice item, where the actual form of $\sigma^\square(S_t)$ depends on the setting we consider, and it will be explained shortly after Eq.(1) below. The natural growth mechanism and propagation mechanism respectively model two different growth processes of an item's popularity in reality. Therefore,

the natural growth mechanism and the propagation mechanism together give the following inductive formulation of the popularity growth of the two items.

$$d_t^p = d_{t-1}^p + \frac{a \cdot d_{t-1}^p}{d_{t-1}^p + d_{t-1}^n}, d_t^n = d_{t-1}^n + \sigma^\square(S_t) + \frac{a \cdot d_{t-1}^n}{d_{t-1}^p + d_{t-1}^n}. \quad (1)$$

In Eq.(1), the notation $\sigma^\square$ needs to be instantiated to a concrete quantity according to the actual setting used. We mainly consider two settings – the overlapping influence (OI) and the non-overlap influence (NI) settings. In the overlapping influence (OI) setting, the influence spread between different rounds are allowed to be overlapped when they are added to the popularity measure. In this case, $\sigma^\square(S)$ is instantiated to $\sigma^{OI}(S)$, and is simply the standard influence spread $\sigma(S)$. In the non-overlapping Influence (NI) setting, the influence spread of S_t in round t cannot overlap with the influence spread in the previous rounds, and thus it is the marginal influence spread of S_t given the seed sets $S_1, \dots, S_{t-1}$ of the previous rounds. Hence, in this case $\sigma^\square(S)$ is instantiated to $\sigma^{NI}(S_t | S_1, \dots, S_{t-1})$, as defined below:

$$\sigma^{NI}(S_t | S_1 \dots S_{t-1}) = \mathbb{E}_{L_1, \dots, L_t} \left[\left| \Gamma(S_t, L_t) \setminus \bigcup_{i=1}^{t-1} \Gamma(S_i, L_i) \right| \right]. \quad (2)$$

where $L_i, i \in [t]$ is the random live-edge graph in round i . When the context is clear, we use $\sigma^{NI}(S_t)$ as a shorthand for $\sigma^{NI}(S_t | S_1 \dots S_{t-1})$.

One remark to Eq.(1) is that both the natural growth and the social network promotion parts are represented in the expectation form. One may formulate them as random variables, and take the expectation in the end. The expectation form is easier to handle, and our empirical evaluation would show that such representation does not lose fidelity in terms of the solution quality.

Another remark concerns the distinction of the natural growth and promotion in our model. The natural growth follows the preferential attachment and is independent of the social network, while the promotion follows the influence diffusion model and is dependent on the social network. One may say that the natural growth may also generate influence diffusion in social networks. The reason we do not include the social network diffusion for the natural growth part of the model is that (a) the natural growth mainly reflects the rich-get-richer effect and models the behavior that people gets the popularity information from online review platforms such

as Yelp or direct offline observations (e.g., by observing the occupancy of the restaurants) and act accordingly, and this part is not related to social network diffusion; and (b) inclusion of the influence diffusion for the natural growth part will further complicate the model, making it less clear in demonstrating the interaction between the preferential attachment for natural growth and the influence diffusion for viral promotion.

Fig. 1 is a numerical example to illustrate our PA-IC model in the OI setting. There are two items in this social network: item p is the popular item with the initial popularity measure $d_0^p = 8$, item n is the novice item with the initial popularity measure $d_0^n = 2$, and the natural growth $a = 5$. All edges in the social network have probability 1. In round 1, according to the PA mechanism, the popularity measure of item p will increase by 4, and the popularity measure of item n will increase by 1. At the same time, the novice item chooses the seed set $S_1 = \{1, 5\}$ to promote their product and the influence spread $\sigma(S_1) = 5$. So the popularity measure of item n will additionally increase by 5. At the end of round 1, $d_1^p = d_0^p + \frac{a \cdot d_0^p}{d_0^p + d_0^n} = 12$, $d_1^n = d_0^n + \frac{a \cdot d_0^n}{d_0^p + d_0^n} + \sigma(S_1) = 8$. In round 2, according to the PA mechanism, the popularity measure of item p will increase by 3, and the popularity measure of item n will increase by 2. In this round, item n chooses the seed set $S_2 = \{9\}$ to promote their product, which generates influence spread $\sigma(S_2) = 5$. So at the end of round 2, $d_2^p = d_1^p = 15$, which means that the novice item has caught up with the popular item.

Finally, the model can be further extended to allow different natural growth count a_t for each round t , or allow the popular item to also have a viral promotion mechanism. We defer the discussion of these model extensions and their impact to our algorithm design to Section 6.

3.2 Popularity Ratio Maximization

We define the *popularity ratio* between the novice and the popular item at the end of round t as $r_t = d_t^n / d_t^p$. Without the promotional mechanism, this ratio would not change. Thus, from the novice item's perspective, it wants to increase this ratio as much as possible and as soon as possible through promotion, but it certainly has budget constraints. Technically, we model this as a popularity ratio maximization problem under a budget constraint. We use the pair (v, t) to denote that the novice item selects v in round t as a seed, i.e., $v \in S_t$, and we use $\mathcal{S} = S_1 \times \{1\} \cup S_2 \times \{2\} \cdots \cup S_T \times \{T\}$ to represent the overall allocation of seeds over T rounds. The novice item has a total budget k , restricting the total number of seeds that it can select, i.e., $|\mathcal{S}| \leq k$. Let the final popularity ratio after T rounds promotion be $r_T(\mathcal{S}) = d_T^n / d_T^p$.

Informally, popularity ratio maximization (PRM) is to find a seed allocation $\mathcal{S}$ with $|\mathcal{S}| \leq k$ for the novice item so that popularity ratio at the end of round T $r_T(\mathcal{S})$ is maximized. In the seed allocation $\mathcal{S}$, the seed set for different round may or may not be allowed to have overlaps, and both may be reasonable depending on the application scenario. Thus, together with the overlapping or non-overlapping influence settings, there are four settings for PRM, as shown in Table 1. Formally, we define the PRM problem below based on these four settings.

Table 1: Four settings of the PRM problem

	Overlap Influence	Non-overlap Influence
Overlap Seeds	OIOS	NIOS
Non-overlap Seeds	OINS	NINS

Definition 1 (Popularity Ratio Maximization). *Given (a) a social network $G = (V, E, p)$, (b) initial popularity measures d_0^n and d_0^p for the novice and popular items respectively, (c) total round number T , (d) natural growth count a , (e) budget k , the task of popularity ratio maximization (PRM) is to find an optimal seed set allocation $\mathcal{S}^*$ for the novice item in one of the four settings depicted in Table 1, such that the total number of seeds does not exceed k , and when the popularity measures evolve according to Eq.(1), the final popularity ratio $r_T(\mathcal{S}^*)$ is maximized. For the OINS setting, it is*

$$\mathcal{S}^* \in \argmax_{\mathcal{S} \subseteq V \times [T], |\mathcal{S}| \leq k, S_t \cap S_{t'} = \emptyset \forall t \neq t'} r_T(\mathcal{S}).$$

Other settings can be similarly formulated.

Note that it may not be wise to use all the budget in the first round since too many seeds could generate redundant activations of the same node, which is only counted once in the popularity measure. On the other hand, spreading the budget evenly across all T round may not be a good choice either, since it would make the popularity growth of the novice item slower and thus harder to catch up with the popular item. Moreover, the objective is to increase the popularity of the novice item in some sense, the problem is still tightly related to the popularity of the novice item, because the PA mechanism links them together. Therefore, PRM is non-trivial both in considering seed allocation across multiple rounds and in considering the relationship with the popular item. Furthermore, PRM needs to consider the impact of different settings. In the following two sections, we will consider two representative settings — overlapping influence and non-overlapping seeds (OINS) and non-overlapping influence and overlapping seeds (NIOS). Other settings can be treated similarly.

PRM can be extended to other variants, and we will consider a number of them in Section 6.

4 RESULTS ON THE OINS SETTING

We study the OINS setting in this section. We first derive the exact formula for the objective function of the PRM problem. We show that the exact objective function is not submodular, implying a rather difficult optimization task. To tackle this problem, we provide a heuristic simplification of the objective function, resulting in a round-weighted influence objective function $\rho_T(\mathcal{S})$, and its corresponding surrogate optimization task called *round-weighted influence maximization (RWIM)*. We show that function $\rho_T(\mathcal{S})$ is monotone and submodular, which allows us to design an efficient approximation algorithm PRM-IMM based on the IMM algorithm [31]. The technical novelty of PRM-IMM includes properly defining the pair-wise reverse reachable set (PW-RR set) to estimate the seed set's influence in the different rounds with different

weights, and calculating the number of PW-RR sets needed to satisfy the approximate ratio. Although our approximation guarantee is on the surrogate RWIM algorithm, our experimental evaluation will demonstrate that it also solves the original PRM problem with the best performance compared to other alternative baselines.

4.1 Objective Function under the OINS Setting

The following lemma states the exact formula for the objective function $r_T^{OI}(S)$ of PRM under the overlapping influence (OI) setting, which is derived recursively.

LEMMA 1. *For the OINS setting, the popularity ratio function at the end of round T is:*

$$r_T^{OI}(S) = (r_0 + 1) \prod_{t=1}^T \left(1 + \frac{\sigma(S_t)}{d_0^n + d_0^p + a \cdot t + \sum_{i=1}^{t-1} \sigma(S_i)} \right) - 1, \quad (3)$$

where $r_0 = d_0^n / d_0^p$ is the initial popularity ratio, $S = \bigcup_{t=1}^T S_t \times \{t\}$.

PROOF. We show Eq.(3) by deriving the recursive formula for r_t , $t \geq 1$. Suppose at the end of round $t-1$ the popularity ratio is r_{t-1} , and the popularity measures for the novice and popular items are d_{t-1}^n and d_{t-1}^p , respectively. By definition, we have

$$r_t + 1 = \frac{d_t^n + d_t^p}{d_t^p} = \frac{d_{t-1}^p + d_{t-1}^n + \sigma^{OI}(S_t) + a}{d_{t-1}^p + a \cdot \frac{d_{t-1}^p}{d_{t-1}^n + d_{t-1}^p}} \quad (4)$$

$$= \frac{d_{t-1}^p + d_{t-1}^n + \sigma^{OI}(S_t) + a}{d_{t-1}^p + d_{t-1}^n + a} \cdot \frac{d_{t-1}^p + d_{t-1}^n}{d_{t-1}^p} \\ = \left(1 + \frac{\sigma^{OI}(S_t)}{d_0^n + d_0^p + a \cdot t + \sum_{i=1}^{t-1} \sigma^{OI}(S_i)} \right) (r_{t-1} + 1), \quad (5)$$

where the second equality in (4) is from Eq.(1), and Eq.(5) uses the fact that $d_{t-1}^p + d_{t-1}^n$ is the total popularity measure of the two items at the end of round $t-1$, which is accumulated from $d_0^n + d_0^p$ by $t-1$ rounds of natural growth ($t \cdot a$) and the promotion effect ($\sum_{i=1}^{t-1} \sigma^{OI}(S_i)$). Eq. (3) can be obtained immediately with the above recursive formula and the fact that $\sigma^{OI}(S_i) = \sigma(S_i)$. $\square$

One feature of Eq. (3) is that the popularity ratio depends on the interplay among the parameters of the novice item (d_0^n), the popular item (d_0^p), the natural growth (a) and the total influence effect $\sigma(S_t)$, which makes the seed selection more complicated in the PRM task than the classical influence maximization task.

A set function $f: 2^V \rightarrow \mathbb{R}$ is monotone if $f(S) \leq f(Q)$ whenever $S \subseteq Q \subseteq V$, and f is submodular if, for any $S \subseteq Q \subseteq V$ and any $x \in V \setminus Q$, $f(S \cup \{x\}) - f(S) \geq f(Q \cup \{x\}) - f(Q)$. Submodularity means that the marginal value of an element with respect to a set S decreases as S grows. Maximizing a monotone and submodular set function with a cardinality constraint can be achieved by a simple greedy algorithm with an approximation ratio of $1 - 1/e$ [24]. Unfortunately, the exact popularity ratio function in Eq. (3) is not submodular, as shown below.

LEMMA 2. *The popularity ratio function $r_T^{OI}(S)$ is monotone but not submodular.*

The proof is omitted, and the non-submodularity is due to the multiplication of submodular functions in Eq.(3).

4.2 Round-Weighted Influence Maximization

Due to the non-submodularity of the popularity ratio function, it is challenging to design a good algorithm for the PRM problem directly. In this section, we simplify the popularity ratio function to obtain a surrogate function and show that this surrogate function is monotone and submodular.

As indicated in the proof sketch of Lemma 2, the complication of the original objective function of Eq.(3) is the series of multiplications, and thus this is the target of our simplification. In particular, our simplification consists of two steps: (a) expanding the multiplication series of Eq.(3) and only keeping the first-order terms; (b) removing the $\sigma(S_1), \dots, \sigma(S_{T-1})$ in the denominator of each term left after step (a). Step (a) decreases the objective value, while step (b) increases the objective value, compensating the effect of step (a). Our experimental evaluation (Section 7.2.2) shows that the surrogate function we develop is reasonably close to the original function in all the test cases. The following is the formula for the surrogate function (ignoring the constant factor $r_0 + 1$ and the additive terms in Eq.(3)):

$$\rho_T^{OI}(S) = \frac{\sigma(S_1)}{d_0^n + d_0^p + a} + \frac{\sigma(S_2)}{d_0^n + d_0^p + 2a} + \dots + \frac{\sigma(S_T)}{d_0^n + d_0^p + T \cdot a}.$$

In the above formula, we can set

$$w_t = \frac{1}{d_0^n + d_0^p + t \cdot a} \quad (6)$$

as the weight for $\sigma(S_t)$ for every round t , and turn the objective function into the following more general objective function:

$$\rho_T^{OI}(S) = \sum_{t=1}^T w_t \cdot \sigma^{OI}(S_t) = \sum_{t=1}^T w_t \cdot \sigma(S_t). \quad (7)$$

Function $\rho_T^{OI}(S)$ means that we need to find an allocation of seeds of T rounds and each round $t \in [T]$ has its own importance weight w_t , which is non-negative and decreasing, i.e., $w_1 \geq w_2 \geq \dots \geq w_T \geq 0$. We call $\rho_T^{OI}(S)$ the *round-weighted influence function*.

The relationship between the popularity ratio function and round-weighted influence function is $r_T^{OI}(S) \approx (1 + \rho_T^{OI}(S))(r_0 + 1) - 1$. The surrogate function is summarized below as the round-weighted influence maximization problem.

Definition 2 (Round-weighted Influence Maximization in the OINS Setting). *Given (a) a social network $G = (V, E, p)$, (b) initial popularity measures d_0^n and d_0^p for the novice and popular items respectively, (c) total round number T , (d) natural growth count a , (e) budget k , (f) the non-negative and non-increasing weight sequence $w_1, w_2, \dots, w_T$, the task of round-weighted influence maximization (RWIM) in the OINS setting is to find an optimal seed set allocation S^* for the novice item under the constraints that the seeds selected for each round are disjoint, and the total number of seeds does not exceed k , such that the round-weighted influence $\rho_T^{OI}(S^*)$ is maximized. That is,*

$$S^* \in \underset{S \subseteq V \times [T], |S| \leq k, S_t \cap S_{t'} = \emptyset \forall t \neq t'}{\operatorname{argmax}} \rho_T^{OI}(S).$$

LEMMA 3. *For any influence graph $G = (V, E, p)$, under the PA-IC model, the round-weighted influence function $\rho_T^{OI}(S)$ defined in Eq.(7) is monotone and submodular.*

4.3 Pair-wise Reverse Reachable Sets

State-of-the-art IM algorithms, including IMM, use the reverse influence sampling (RIS) approach governed by reverse reachable (RR) sets. An RR set is a random set of nodes sampled from the graph by (a) first selecting a node v uniformly at random from the graph, and (b) simulating the reverse propagation of the model (e.g., IC model) and adding all visited nodes into the RR set. The main property of a random RR set R is that: influence spread $\sigma(S) = N \cdot \mathbb{E}[\mathbb{I}\{S \cap R \neq \emptyset\}]$ for any seed set S , where $\mathbb{I}$ is the indicator function, N is the number of nodes in graph G . After finding a large enough number of RR sets, the original influence maximization problem is turned into a k -max coverage problem — finding a set of k nodes that covers the most number of RR sets, where a set S covers an RR set R means that $S \cap R \neq \emptyset$. All RIS algorithms have the following two steps: 1. Generate a sufficiently large set of random RR sets, called Sampling step. 2. Find k nodes that cover the most number of RR sets, called NodeSelection step.

We now introduce our adaptation of the RR set to the PRM setting, particularly, the pair-wise reverse reachable (PW-RR) set.

4.3.1 PW-RR set and its generation process. Since the RR set is not designed to estimate the influence of nodes in different rounds, we define the pair-wise reverse reachable (PW-RR) set for the PRM problem and denote it as $R^{(t)}$. PW-RR can distinguish between different rounds of RR set, and it also corresponds to different weights for different rounds of RR sets so that the round-weighted influence of nodes in different rounds can be calculated. A (random) pair-wise RR set $R^{(t)}$ is an RR set R rooted at a node picked uniformly at random from V , and t is picked uniformly at random from $[T]$, and $R^{(t)} = R \times \{t\}$.

4.3.2 Property of PW-RR set. The main property of random PW-RR set $R^{(t)}$'s is that we can use them to estimate the round-weighted influence spread $\rho_T^{OI}(S)$.

LEMMA 4. For any set of node-round pairs S , we have

$$\rho_T^{OI}(S) = N \cdot T \cdot \mathbb{E} \left[w_t \cdot \mathbb{I} \left[S \cap R^{(t)} \neq \emptyset \right] \right], \quad (8)$$

where $R^{(t)}$ is a random PW-RR set, and the expectation is taken from the randomness of (a) the root of a PW-RR set $R^{(t)}$ uniformly chosen at random, (b) the randomly generated RR set from the root, and (c) round number t uniformly chosen at random.

PROOF. The lemma is shown by the following:

$$\mathbb{E} \left[w_t \cdot \mathbb{I} \left[S \cap R^{(t)} \neq \emptyset \right] \right] = \frac{1}{T} \sum_{t=1}^T w_t \cdot \mathbb{E} \left[\mathbb{I} \{S \cap R \neq \emptyset\} \right] \quad (9)$$

$$= \frac{1}{T} \sum_{t=1}^T w_t \cdot \frac{1}{N} \sigma(S_t) = \frac{1}{N \cdot T} \rho_T^{OI}(S), \quad (10)$$

where the R in Eq.(9) is a random RR set as defined at the beginning of this section, the first equality in (10) is by the property of the RR set as given at the beginning of this section, and the second equality in (10) is by Eq.(7). $\square$

Lemma 4 reflects the overlapping influence (OI), and it is clearly seen in Eq.(10) where influence spread of different rounds are summed up, allowing overlapping influence.

Algorithm 1 PRM-NodeSelection-OINS

Input: PW-RR sets $\mathcal{R}$, budget k , weights $(w_t)_{t \in [T]}$ (Eq.(6))

Output: seed set $\hat{S}^g$

```

1:  $RR[(v, t)] = \{R \in \mathcal{R} | (v, t) \in R\}$ ,  $c[(v, t)] = w_t \cdot |RR[(v, t)]|$ ,
    $\forall (v, t) \in V \times [T]$  /*  $RR[(v, t)]$ 's and  $c[(v, t)]$ 's can be constructed during the generation of  $\mathcal{R}^*$  */
2: for all  $R \in \mathcal{R}$ ,  $t \in [T]$ ,  $covered[R] = false$ 
3:  $S = \emptyset$  /* the element of  $S$  is  $(v, t)$  */
4: for  $i = 1$  to  $k$  do
5:    $(v, t) = \operatorname{argmax}_{(v, t) \in (V \setminus S_{node}) \times [k]} c[(v, t)]$ 
6:   /*  $S_{node}$  is the node set of  $S^*$  */
7:    $S = S \cup \{(v, t)\}$ 
8:   for  $R \in RR[(v, t)] \wedge covered[R] == false$  do
9:      $covered[R] = true$ 
10:    for  $(u, t) \in R \wedge u \neq v$  do
11:       $c[(u, t)] = c[(u, t)] - w_t$ 
12:    end for
13:  end for
14: end for
15: return  $\hat{S}^g = S$ 
```

4.4 Popularity Ratio Maximization IMM

We could base the key formula of Eq.(8) to estimate the overlapping influence spread $\rho_T^{OI}(S)$. Let $\mathcal{R}$ be a collection of θ PW-RR sets we generated independently, $\mathcal{R} = \{R_1^{(t_1)}, R_2^{(t_2)}, \dots, R_\theta^{(t_\theta)}\}$. Define random variable $Y_i^{\mathcal{R}}(S) = w_{t_i} \cdot \mathbb{I} \left[S \cap R_i^{(t_i)} \neq \emptyset \right]$. We can estimate $\rho_T^{OI}(S)$ using the following estimator $\hat{\rho}_T^{OI}(S, \mathcal{R})$:

$$\hat{\rho}_T^{OI}(S, \mathcal{R}) = \frac{N \cdot T}{\theta} \sum_{i=1}^{\theta} Y_i^{\mathcal{R}}(S).$$

Estimator $\hat{\rho}_T^{OI}(S, \mathcal{R})$ can be viewed as a weighted coverage function on a bipartite graph with $V \times [T]$ on the one side and PW-RR sets $\mathcal{R}$ on the other side.

Our PRM-IMM algorithm follows the IMM structure [31] and consists of two components: PRM-NodeSelection for selecting top k pairs from the above bipartite graph, and PRM-Sampling for generating enough PW-RR set samples. We now explain the main adaption of these two components to our PRM settings.

4.4.1 PRM-NodeSelection-OINS. Algorithm 1 provides the pseudocode for the PRM-NodeSelection under the OINS setting, which is patterned from the similar algorithms in [7, 31]. It mainly implements a greedy selection of k pairs to cover the PW-RR sets. Several points worth mentioning are: (a) the greedy algorithm works on PW-RR sets for (v, t) pairs, as described in Section 4.3; (b) the overlapping influence (OI) aspect is reflected in the PW-RR set design and Lemma 4; and (c) the non-overlapping seed (NS) aspect is reflected in line 5, where nodes already selected, S_{node} , are removed from further consideration in seed selection.

Due to the NS requirement, the feasible solutions for the bipartite coverage problem actually form a partition matroid.¹ The greedy

¹A matroid $(E, \mathcal{I})$ on a finite ground element set E is a collection $\mathcal{I}$ of subsets of E , each called an independent set, with the following two properties: (a) a subset of an independent set is still an independent set; and (b) for two independent sets $I_1, I_2 \in \mathcal{I}$,

algorithm on the partition matroid gives a $1/2$ approximation of the optimal solution.²

4.4.2 PRM-Sampling. The sampling procedure also follows the IMM structure (Algorithm 2), in that it uses iterative halving to estimate a lower bound LB of the optimal solution OPT , and then use LB to determine the final number of PW-RR sets needed. The parameters α, β are defined below:

$$\alpha = \sqrt{l \ln N + \ln 4}, \beta = \sqrt{\frac{1}{2} \cdot \left(\ln \binom{N}{k} + \alpha^2 + k \ln T \right)}. \quad (11)$$

There are a few differences compared to the classical IMM. First, the θ_i (line 4) and $\tilde{\theta}$ (line 12) have additional coefficients w_1 and T in the numerator. Factor T is because we allocate seeds in T rounds. Factor w_1 is technically because random variable $Y_i^R(\mathcal{S})/w_1$ is bounded within $[0, 1]$, so that we can apply the Chernoff bound on it to obtain the desired result. Second, the parameter β also adds an extra $k \ln T$, because in our problem, the number of possible solutions is upper bounded by $\binom{N}{k} \cdot T^k$, which means that k nodes are selected among N nodes and each selected node may be placed in any of the T rounds. Of course, our approximation guarantee is also changed from $1 - 1/e - \varepsilon$ to $1/2 - \varepsilon$, because we are dealing with a more general partition matroid constraint. The regeneration of PW-RR sets in line 13 ensures that all PW-RR sets in $\mathcal{R}$ are mutually independent [6].

The following theorem summarizes the correctness and time complexity of our algorithm PRM-IMM.

THEOREM 5. *Let $\hat{\mathcal{S}}^g$ be the output of PRM-IMM-OINS and $\mathcal{S}^*$ be the optimal solution of round-weighted influence maximization problem. We have that with the probability at least $(1 - \frac{1}{N^l})$,*

$$\rho_T^{OI}(\hat{\mathcal{S}}^g) \geq \left(\frac{1}{2} - \varepsilon \right) \rho_T^{OI}(\mathcal{S}^*),$$

where $\varepsilon > 0, l > 0$. The expected running time of PRM-IMM-OINS is $O((k+l)(M+N)T \log(N \cdot T)/\varepsilon^2)$.

The proof of the theorem follows the analysis structure of IMM [31], and utilizes the key result in Lemma 4. We can see that PRM-IMM-OINS provides the approximation guarantee while still runs in time near linear to the graph size. Compared to the original IMM, the running time is increased by a factor of T , caused by the spreading of search of seeds across T rounds rather than in one round.

5 RESULTS ON THE NIOS SETTING

We now turn to the non-overlapping influence with overlapping seeds (NIOS) setting. The NI setting causes important changes to the objective function and the algorithm design, as we will describe in this section.

if $|I_1| < |I_2|$, then there is at least one element $e \in I_2 \setminus I_1$ such that $I_1 \cup \{e\}$ is also an independent set. A partition matroid is a matroid where the ground set E is partitioned into disjoint sets $E_1, E_2, \dots, E_s$, and each E_i has a capacity constraint b_i such that for all $I \in \mathcal{I}$, $|I \cap E_i| \leq b_i$.

²We are aware that there is a polynomial-time algorithm that achieves $(1 - 1/e)$ -approximation for submodular maximization under a matroid constraint [5], but the algorithm is highly inefficient, and thus we choose the simple greedy algorithm for our implementation.

Algorithm 2 PRM-IMM-OINS algorithm

Input: directed graph $G = (V, E)$, round number T , budget k , weights $(w_t)_{t \in [T]}$ (Eq.(6)), approximate ratio parameter ε , Error probability parameter l .

Output: $\hat{\mathcal{S}}^g$

```

1:  $\mathcal{R} = \emptyset, LB = w_1, \varepsilon' = \sqrt{2}, \theta_0 = 0$ 
2: for  $i = 1$  to  $\lfloor \log_2 N \rfloor - 1$  do
3:    $x_i = \frac{\sum_{t=1}^T w_t \cdot N}{2^i}$ 
4:    $\theta_i = \left\lceil \frac{w_1 \cdot N \cdot T (2 + \frac{2}{3} \varepsilon') (\ln \log_2 N + 2\beta^2)}{\varepsilon'^2 x_i} \right\rceil$ 
5:   independently generate  $\theta_i - \theta_{i-1}$  PW-RR sets into  $\mathcal{R}$ :
6:    $\mathcal{S} = \text{PRM-NodeSelection-OINS}(\mathcal{R}, k, (w_t)_{t \in [T]})$ 
7:   if  $\hat{\rho}_T(\mathcal{S}, \mathcal{R}) \geq (1 + \varepsilon') x_i$  then
8:      $LB = \frac{\hat{\rho}_T(\mathcal{S}, \mathcal{R})}{1 + \varepsilon'}$ 
9:     break
10:  end if
11: end for
12:  $\tilde{\theta} = \left\lceil \frac{2w_1 \cdot N \cdot T (\frac{1}{2} \alpha + \beta)^2}{LB \cdot \varepsilon^2} \right\rceil$   $\alpha$  and  $\beta$  is defined in Eq.(11).*/
13: clear the  $\mathcal{R}$ , and regenerate  $\tilde{\theta}$  PW-RR sets into  $\mathcal{R}$ 
14:  $\hat{\mathcal{S}}^g = \text{PRM-NodeSelection-OINS}(\mathcal{R}, k, (w_t)_{t \in [T]})$ 
15: return  $\hat{\mathcal{S}}^g$ 
```

5.1 Objective Function under the NIOS Setting

Similar to Eq.(3) for the OINS setting, we have

$$r_T^{NI}(\mathcal{S}) = (r_0 + 1) \prod_{t=1}^T \left(1 + \frac{\sigma^{NI}(\mathcal{S}_t)}{d_0^n + d_0^p + a \cdot t + \sum_{i=1}^{t-1} \sigma^{NI}(\mathcal{S}_i)} \right) - 1. \quad (12)$$

The main difference is that it uses the marginal influence $\sigma^{NI}(\mathcal{S}_t)$ as defined in Eq.(2), because influence cannot be overlapping under NIOS. With a similar heuristic simplification, we obtain the round-weighted influence function with w_t defined in (6):

$$\rho_T^{NI}(\mathcal{S}) = \sum_{t=1}^T w_t \cdot \sigma^{NI}(\mathcal{S}_t). \quad (13)$$

One can show that $\rho_T^{NI}(\mathcal{S})$ has the following equivalent form:

$$\rho_T^{NI}(\mathcal{S}) = \sum_{v \in V} \mathbb{E}_{L_1, \dots, L_T} \left[\max_{t \in [T]} w_t \cdot \mathbb{I}\{v \in \Gamma(\mathcal{S}_t, L_t)\} \right]. \quad (14)$$

This is because for any node $v \in V$, it may be influenced by some pair $(u, t) \in \mathcal{S}$ in round t , and when multiple seed-round pairs occur, the one with the largest weight among these pairs is taken to calculate the final influence to v . Based on Eq.(14), one can show that $\rho_T^{NI}(\mathcal{S})$ is monotone and submodular.

5.2 Efficient Algorithm for NIOS setting

The PRM-IMM-NIOS algorithm follows the same structure as the PRM-IMM-OINS algorithm, but due to the non-overlapping influence requirement and the round-based weights nature, the RR set and the node selection procedure has significant difference, as we now explain.

For the RR set, for each randomly selected root $v \in V$, we need to generate T RR sets rooted at v , denoted as $R_{v,t}, t \in [T]$ in pair

Algorithm 3 PRM-NodeSelection-NIOS

Input: the set of MR-RR sets $\mathcal{R}$, seeds budget k , weights $(w_t)_{t \in [T]}$ (Eq.(6)). particularly, the value of w_{T+1} is 0.

Output: k size seed set $\hat{\mathcal{S}}^g$.

```

1:  $RR[(v, t)] = \{R \in \mathcal{R} | (v, t) \in R\}$ ,  $c[(v, t)] = w_t \cdot |RR[(v, t)]|$ ,
    $\forall (v, t) \in V \times [T]$  /*  $RR[(v, t)]$ 's and  $c[(v, t)]$ 's can be constructed during the generation of  $\mathcal{R}$  */
2: for all  $R \in \mathcal{R}$ ,  $t_R = T + 1$ ;  $w_{T+1} = 0$ 
3:  $\mathcal{S} = \emptyset$  /* the element of  $\mathcal{S}$  is  $(v, t)$  */
4: for  $i = 1$  to  $k$  do
5:    $(v, t) = \operatorname{argmax}_{(v, t) \in V \times [k] \setminus \mathcal{S}} c[(v, t)]$ 
6:    $\mathcal{S} = \mathcal{S} \cup (v, t)$ 
7:   for  $R \in RR[(v, t)] \wedge t < t_R$  do
8:     for  $(u, t') \in R \wedge (u, t') \neq (v, t) \wedge t' < t_R$  do
9:       if  $t' \geq t$  then
10:         $c[(u, t')] = c[(u, t')] - (w_{t'} - w_{t_R})$ 
11:       else
12:         $c[(u, t')] = c[(u, t')] - (w_t - w_{t_R})$ 
13:       end if
14:     end for
15:    $t_R = t$ 
16:   end for
17: end for
18: return  $\hat{\mathcal{S}}^g = \mathcal{S}$ 

```

notation. The multi-round RR set (MR-RR set) is the union of all T RR sets, as $R_v = \bigcup_{t=1}^T R_{v,t}$. This is because in the NI setting a node v is only influenced once in all T rounds. The MR-RR set is similar to the one used in [28], but here different rounds have different weights, leading to more complicated node selection procedure.

In the PRM-NodeSelection-NIOS procedure (Algorithm 3), we maintain a variable t_R for each MR-RR set R , and it denotes the lowest round number t in which the root v_R of R has been influenced by some already selected seed nodes. Initially, $t_R = T + 1$, since no seed has been selected and thus no node is influenced. We also use variable $c[(u, t)]$ to store the remaining marginal influence of pair (u, t) on all MR-RR sets. Variables t_R and $c[(u, t)]$ are used to keep track of the marginal influence to v_R when some additional pair (v, t) is selected as a seed (lines 7–16): when $t < t_R$, this means that the root v_R is now influenced at the lowest round t , and thus for any other pair $(u, t') \in R$ with $t' < t_R$, if $t' \geq t$, it no longer generates any marginal influence to v_R (the marginal influence is reduced by $w_{t'} - w_{t_R}$, line 10); and if $t' < t$, the remaining marginal influence (u, t') could generate to v_R through R is $w_{t'} - w_t$ (the marginal influence is reduced by $w_t - w_{t_R}$, line 12). This makes this node selection procedure more complicated than the OINS setting. Note that in line 5, we select the next seed from $V \times [k] \setminus \mathcal{S}$, reflecting that we allow overlapping seeds in different rounds.

In the sampling part, similar to OINS setting, we can iteratively guess the lower bound x_i by halving and verifying through NodeSelection. We only need to modify parameter β to $\sqrt{\frac{1}{2} \cdot \left(\ln \binom{NT}{k} + \alpha^2\right)}$, and the $\binom{NT}{k}$ is due to selecting k pairs from all NT pairs in the overlapping seeds setting. All other aspects of sampling part remains the same, and thus we ignore the pseudocode here.

For the other two problems mentioned in Section 3.2, NINS and OIOS, we can plug in different components in OINS and NIOS to achieve the goal, and we omit the details here.

6 OTHER VARIANTS OF PRM PROBLEM

In this section, we show that our object function analysis and algorithm design are robust in that they can be adapted to a number of variants of the PRM problem studied in the previous sections.

6.1 Per-round Budget

In the PRM definition (Definition 1), the budget k is over all T rounds. One variant is that every round has its own budget k , similar to [28]. It also has four settings as in Table 1. Among them, the OIOS setting is the same as solving the traditional IM problem and then using the same seed set for all rounds. Other settings can be solved following the similar technique as the overall budget case, but when selecting next node-round pair as a seed, it needs to respect the per-round budget rather than the overall budget.

6.2 Variable Natural Growth Count

We can relax the assumption that the natural growth count a is constant for all rounds to allow a natural growth vector $\mathbf{a} = [a_1, a_2, \dots, a_T]$ with a_t denoting the natural customer count in round t . Then, similar to the derivation process in Section 4.1, we can obtain the formula for $r_T^{OI}(\mathcal{S})$ as follows:

$$r_T^{OI}(\mathcal{S}) = (r_0 + 1) \prod_{t=1}^T \left(1 + \frac{\sigma(S_t)}{d_0^n + d_0^p + \sum_{i=1}^t a_i + \sum_{i=1}^{t-1} \sigma(S_i)} \right) - 1.$$

The derivation of the surrogate function $\rho_T^{OI}(\mathcal{S})$ and the rest algorithm design is the same, and we just need to redefine the weight $w_t = 1/(d_0^n + d_0^p + \sum_{i=1}^t a_i)$.

6.3 Popular Item Engages in Promotion

Our PRM problem considers passive popular item that does not engage in promotion. When popular item also engages in promotion, the situation is more complicated. Here we assume that the promotion of the popular item in all rounds are known, and the promotional result on round t is given as p_t , $t \in [T]$. The case of unknown popular item promotions would lead the problem into the game-theoretic setting, which is beyond the scope of this paper. When $p_1, \dots, p_T$ are known, in the OINS setting, the close form of the objective function $r_T^{OI}(\mathcal{S})$ is not easily derived as in Lemma 1. Instead we can obtain its upper and lower bound as follows.

$$r_T < (r_0 + 1) \prod_{t=1}^T \left(1 + \frac{\sigma(S_t)}{d_0^n + d_0^p + at + \sum_{i=1}^{t-1} (\sigma(S_i) + p_i)} \right) - 1, \quad (15)$$

$$r_T > (r_0 + 1) \prod_{t=1}^T \left(1 + \frac{\sigma(S_t) - p_t}{d_0^n + d_0^p + at + \sum_{i=1}^{t-1} (\sigma(S_i) + p_i) + p_t} \right) - 1. \quad (16)$$

Eq.(15) can be viewed as treating p_t as the additional natural growth in round t , and thus disfavor of the popular item and in favor of the novice item. Eq.(16) works when $d_p^t \geq d_n^t$, i.e. popular item is

Table 2: Datasets

	DM	LastFM	FX01/02	HEPT	Yelp	DBLP	LJ
nodes	0.6K	1.8K	29.3K	27.7K	110K	650K	4800K
edges	3.3K	12.7K	212.6K	352.8K	950K	1900K	42800K
avg.degree	4.96	13.44	7.24	12.70	15.92	3.04	8.83

Table 3: Parameter Settings

	DM	LastFM	FX01	FX02	HEPT	Yelp	DBLP	LJ
r_0	0.25	0.067	0.13	0.067	0.25	0.02	0.025	0.025
d_0	900	1600	1700	1600	5000	51000	20500	205000
a	10	100	50	100	150	2000	2500	25000

still the majority in popularity. These bounds lead to two weight settings:

$$\bar{w}_t = \frac{1}{d_0^n + d_0^p + t \cdot a + \sum_{i=1}^t p_i}, \underline{w}_t = \frac{1}{d_0^n + d_0^p + t \cdot a + \sum_{i=1}^t p_i + p_t}.$$

Then following the sandwich approximation strategy [23], we can apply the two sets of weights to the corresponding PRM-IMM algorithm, and obtain two seed allocations, and compare their results to find the better one. The approximation ratio for the surrogate objective would have an extra factor of $\max_{t \in T} \underline{w}_t / \bar{w}_t$.

6.4 Two Other Optimization Objectives

Beyond the fundamental problem of maximizing the popularity ratio, it is natural to investigate two other variants of the PRM problem in surpassing the competitor: minimizing seed budget and minimizing the number of rounds needed.

- **Seed Minimization:** Given a round budget T , minimize the number of seeds in $\mathcal{S}$ such that the final popularity ratio is at least 1, i.e., $r_T(\mathcal{S}) \geq 1$.
- **Round Minimization:** Given a seed budget k , minimize the round number T such that the final popularity ratio is at least 1, i.e., $r_T(\mathcal{S}) \geq 1$.

Note that $r_T(\mathcal{S}) \geq 1$ means that the novice item has successfully caught up or surpassed the popular item at the end of the promotional period. For the seed minimization problem, we first find the range of minimum budget by doubling the budget from 1, and then determine the minimum budget that would make $r_T \geq 1$ by a binary search on the budget, and for each budget tried, we use the PRM-IMM algorithms developed. Same binary search idea can be applied to the round minimization problem.

7 EXPERIMENTS

In this section, we empirically validate our algorithms on eight real-world networks.

7.1 Experiment Setup

7.1.1 Data description. We apply eight real-world network datasets in our experiments, with basic statistics shown in Table 2. The DM

dataset is a collaboration network where nodes represent the data-mining researchers and edges represent co-authorships [29]. The FX01/02 datasets are from the Flixster, a network of American social movie discovery services [2]. Each node is a user and a directed edge from node u to v is formed if v rates one movie shortly after u does so on the same movie. FX01/02 represent two networks on two different movie topics [2]. The NetHEPT dataset is an academic collaboration network from the “High Energy Physics Theory” section of arXiv from 1991 to 2003, where nodes represent the authors and each edge represents one paper co-authored by two nodes [9]. LastFM is a UK-based music site that focuses on online radio and music communities³. We use the friend relationship on LastFM to build a user relationship social network. The Yelp dataset is released by yelp officially⁴. Yelp.com is the largest review site in the United States. We use the friend relationship data in Yelp to build a user’s social network. DBLP is another academic collaboration network extracted from the online archive DBLP (dblp.uni-trier.de) and used for influence studies [33]. LiveJournal, denoted as LJ, is a free online community with almost 10 million members; a significant fraction of these members are highly active. It is obtained from Stanford’s SNAP project [19].

7.1.2 Metrics. We consider the following three metrics, reflecting what one could consider in practice:

- **Popularity ratio.** The popularity ratio at the end of promotions.
- **Surpass budget.** Given a fixed round T , the minimum budget needed for the novice item to surpass the popular item.
- **Surpass time.** Given a fixed budget k , the minimum round number when the novice item surpasses the popular item.

7.1.3 Parameter setup and reproducibility. For DM and FX datasets, we use the edge weights learned in their respective studies [2, 29]. For the remaining datasets, we use the following “Weighted Cascade” policy adopted in [17] and often used in other influence maximization studies (e.g. [10, 31, 33]): for each edge (u, v) , $p(u, v)$ is set to be $\frac{1}{N_{in}(v)}$, where $N_{in}(v)$ is the indegree of v . Every PRM instance is also determined by the parameters r_0, d_0, a . We set proper parameters for each dataset, as shown in Table 3, to reflect different scenarios and match with the network size. We also vary these parameters and study their effects (Fig.4 and Fig.5).

Following [31], we set the error probability parameter $l = 1$, approximation ratio parameter $\epsilon = 0.1$ in PRM-IMM.

For implementation details see the source code in this repository⁵

7.1.4 Baselines. There is no existing baseline for the newly proposed PRM problem. Thus, we compare with the following reasonable heuristic baselines.

- **OneShot-IMM:** The largest k nodes are picked by the IMM algorithm, and all are placed in round 1.
- **Uni-IMM-NS/Uni-IMM-OS:** The NS version selects the largest k nodes by IMM and places them in all rounds separately with equal size in the order of the greedy selection. The OS version simply repeats the first round seeds in later rounds (for NIOS tests).

³<http://www.last.fm/>

⁴<http://www.yelp.com>

⁵<https://anonymous.4open.science/r/PRM-37A9>

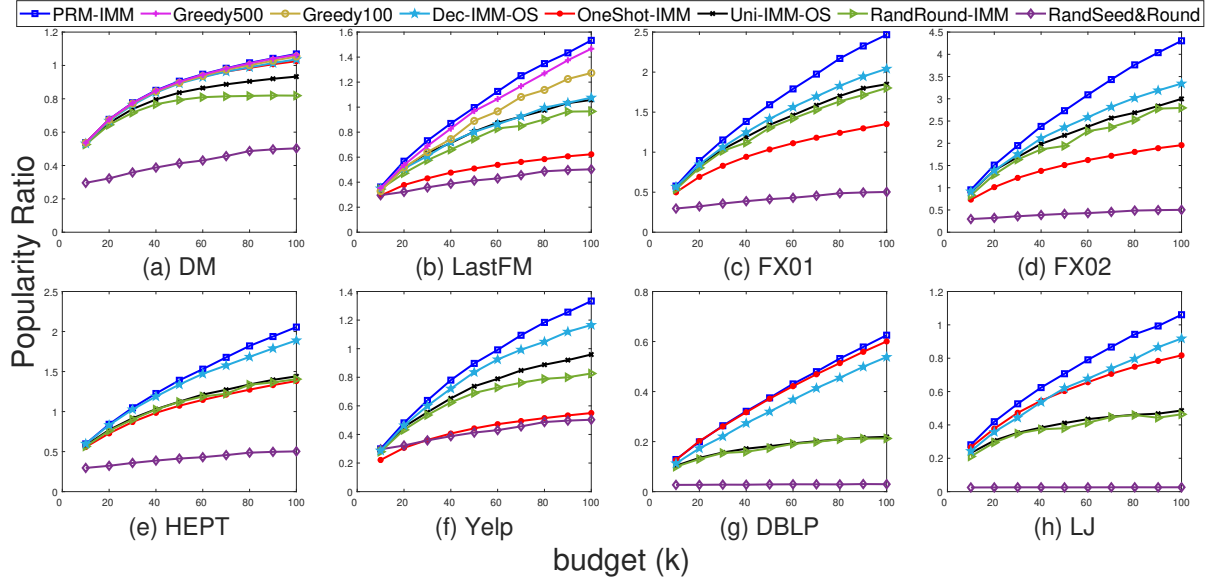


Figure 2: The popularity ratio vs. budgets for different algorithms in OINS setting.

- **RandRound-IMM**: Pick the largest k nodes by the IMM algorithm and put them into T rounds randomly.
- **RandSeed&Round**: Randomly select k nodes and randomly place these nodes in each round.
- **Dec-IMM-NS/Dec-IMM-OS**: The NS version picks the largest k nodes by IMM and places seed nodes in descending order. The method of generating the decreasing sequence is to use one-fifth of the budget in the first round and one-fifth of the remaining budget in each subsequent round. The OS version repeats the seeds in the first round with the same budget as the NS version.
- **Greedy**: Directly apply the greedy algorithm, using simulations to estimate marginal gains and do not use the RIS approach.

The first five baselines are all heuristics with no theoretical guarantee. Greedy algorithm has the theoretical guarantee as PRM-IMM, but it is very slow in running simulations, and we will only show its result on the two small datasets DM and LastFM.

7.1.5 Platform. We implement all algorithms in Visual C++, compiled in Visual Studio 2019, and run our tests on a computer with 3.6GHz Intel(R) Core(TM) i9-9900K CPU (16 cores), 128G memory, and Windows 10 professional (64 bits).

7.2 Results

7.2.1 OINS main results. Fig. 2 shows the result of popularity ratio vs. budget on all datasets and all algorithms in the OINS setting. The results are the average over 10 independent runs. Note that the y-axis corresponds to the original objective function (Eq. (3)), not the surrogate function (Eq. (7)). It is obvious that PRM-IMM outperforms the baseline algorithms with a large margin in many cases. More specifically, RandomSeed&Round is much worse than other algorithms, showing that randomly giving out promotions should be avoided if possible. Uniform-IMM and RandomRound-IMM have similar performance since both spread the seeds more

Table 4: Running Time (OINS setting, $T = 100$, $k = 100$, results in seconds, average over 5 runs)

	IMM	PRM-IMM	Greedy100	Greedy500
DM	0.067	13.7	397.8	3489
LastFM	1.24	82.86	11976	53681
FX01	4.07	437	-	-
FX02	3.42	432	-	-
HEPT	2.39	399	-	-
Yelp	44.36	2959	-	-
DBLP	47.46	1176	-	-
LJ	371	15026	-	-

or less evenly among the T rounds, and in many cases such even spreading results in poor performance. OneShot-IMM is at the opposite extreme, only allocating seeds in the first round. It has unstable performance, and in most cases it performs poorly, except for the DM and DBLP datasets. Decreasing-IMM in general performs better than the OneShot/Uniform/RandomRound versions, but it still has a clear gap with our algorithm PRM-IMM. This is because all these heuristics cannot adjust to different network structure and diffusion characteristics of the networks (such as whether important seeds are likely to have overlapping influence), while our PRM-IMM automatically balances the early promotion and the influence redundancy among seeds.

Greedy should perform equally well as PRM-IMM, since PRM-IMM is just a more efficient way of implementing Greedy. But Greedy is very time consuming — as shown in Table 4, Greedy with 100 and 500 simulations per estimation is already several orders of magnitude slower than PRM-IMM, and it takes Greedy500 almost 15 hours to run on the 1.8K node LastFM network. Thus Greedy

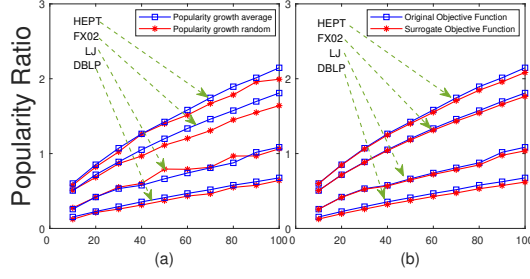


Figure 3: Model justification. (a) Comparing between the random growth (red) and average growth (blue); (b) Comparing between the original objective function and the surrogate objective function.

is not scalable and we cannot run it on other datasets except for DM and LastFM. Even for LastFM, Fig. 2(e) shows that Greedy500 and Greedy100 are not as competitive as PRM-IMM, due to the insufficient number of simulations.

Table 4 also shows the running time of PRM-IMM is around 100 times slower than the pure IMM algorithm (except for DBLP and LJ), which aligns with our theoretical analysis showing that our running time has an extra T factor. From the two large datasets DBLP and LJ, we reduce its round input T to save time, since the results from smaller budgets show that only the first few rounds get allocated for seeds.

Overall, our OINS results clearly show that our PRM-IMM algorithm outperforms all simple heuristics, while it runs much more efficiently than the greedy algorithm. PRM-IMM achieves a good balance between quality and efficiency.

7.2.2 Model justification. We now justify two choices we made in our model formulation. First, Eq.(1) is based on the average popularity growth in each round rather than the random popularity growth, as we discussed in Section 3.1. We compare the effect of the average growth with the random growth, using the seed allocation selected by PRM-IMM with the budget ranging from 10 to 100. Fig. 3(a) shows that the results of the two effects are very close in four datasets (the other four have similar results), justifying our use of the average effect in Eq.(1).

Next, we compare the original objective function of Eq. (3) with the surrogate round-weighted influence function of Eq. (7). We use seed allocations selected by PRM-IMM for this comparison. Fig. 3(b) shows that the two functions agree with each other very well, with average derivation less than 5% for four datasets (and the other four have similar results). Hence, using the surrogate function is a reasonable choice.

7.2.3 Parameter analysis in PRM. We now analyze the effect of the initial total popularity measure d_0 and natural growth count a on the PRM-IMM algorithm. We run PRM-IMM to select 100 seeds for this analysis. Fig. 4 shows the result for FX01, and results on other datasets are similar. Fig. 4(a) shows that when a increases, the popularity ratio decreases, indicating that the larger the natural growth of the popularity measure, the more difficult it is for the novice item to catch up with the popular item. Fig. 4(b) also shows the decreasing trend with respect to d_0 . This is because when d_0 is

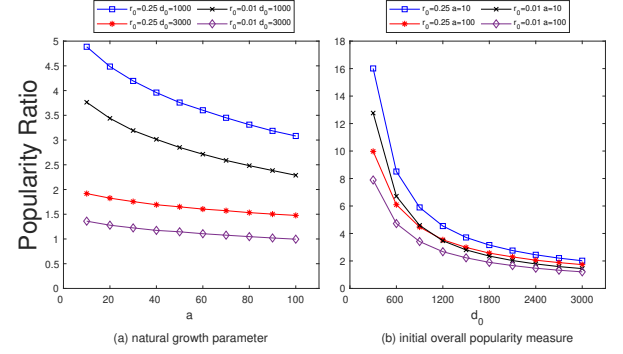


Figure 4: Parameter analysis for PRM under FX01.

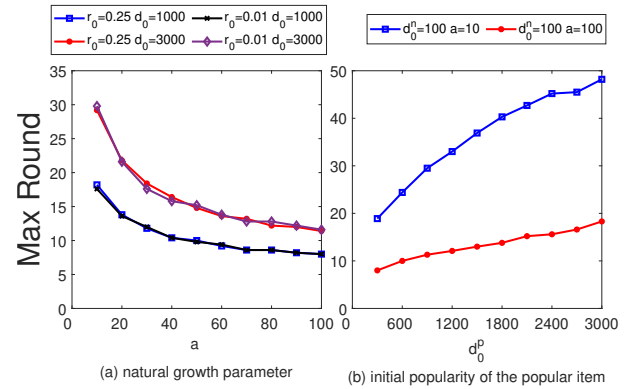


Figure 5: Different parameter's impact on seeds allocation, in the FX01 dataset.

Table 5: Running Time (NIOS setting, $T = 100$, $k = 100$, results in seconds, average over 5 runs)

	IMM	PRM-IMM	greedy-100	greedy-500
DM	0.067	342.8	1274.8	3489
LastFM	1.24	1258	51034	249656
FX01	4.07	4396	-	-
FX02	3.42	6163	-	-
HEPT	2.39	8056	-	-
Yelp	44.36	39582	-	-
DBLP	47.46	1395	-	-
LJ	371	76396	-	-

larger while r_0 remains the same, the gap between the popular and novice item becomes larger and it is more difficult for the novice item to catch up.

Among the parameters, a and d_0^p (initial popularity of the popular item) are not directly related to the novice item. Thus, we further investigate how a and d_0^p would affect the seed allocation choice of the novice item in Fig. 5. Y-axis “max round” means the maximum number of rounds used in the allocation. Fig. 5 shows that when a increases, the max round decreases, while when d_0^p increases,

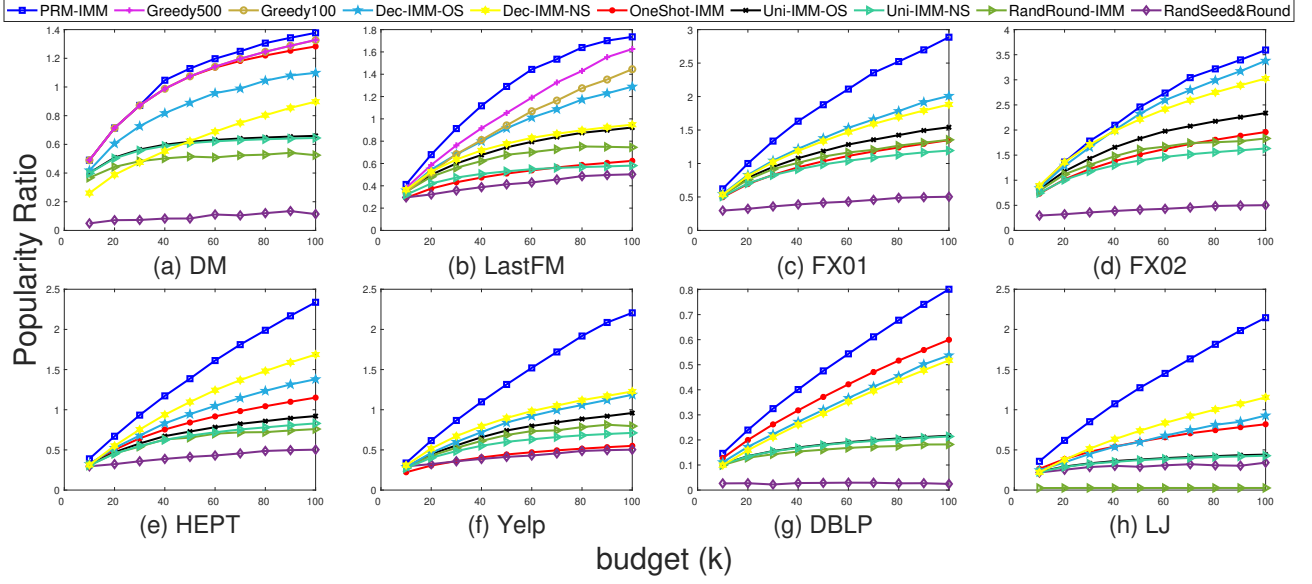


Figure 6: The popularity ratio vs. budgets for different algorithms in NIOS setting

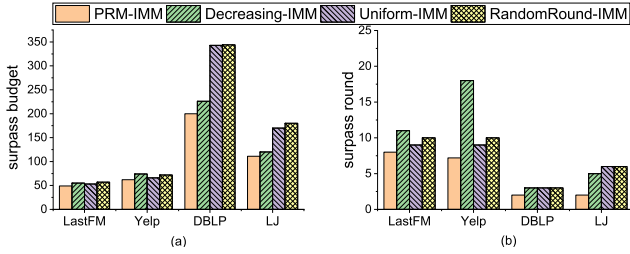


Figure 7: (a) Fix round $T = 20$, find the smallest budget to make the popularity ratio exceed 1. (b) Fix budget (LastFM, $k = 55$; Yelp, $k = 75$; DBLP, $k = 230$; LJ, $k = 125$), make the popularity ratio exceed 1 in as few rounds as possible.

the max round also increases. This shows the interesting adaptive strategy of PRM-IMM: when every round the natural growth count increases, the algorithm tries to put more seeds earlier, otherwise natural growth in each round will cause more people going to the popular item and making it harder to catch up; but if only the initial popularity of the popular item increases, the algorithm can use more rounds to achieve a better catch-up result in the end.

7.2.4 Results on NIOS. The popularity ratio and running time results for NIOS are shown in Fig. 6 and Table 5, respectively. For this test, we further include baselines Uni-IMM-OS and Dec-IMM-OS to consider seed overlaps. Overall, the results are similar to those of OINS, and our PRM-IMM algorithm always performs the best among all baselines, and it runs significantly faster than the simple greedy algorithm. The performance gap between our algorithm and other baselines is even larger in many cases compared to OINS. This is perhaps because the NIOS requirement demands more sophisticated seed allocation. For example, we notice that seed overlaps vary depending on the networks. The OS specific baselines in many

cases perform not as well as their NS counterparts, indicating that it is not easy to adjust proper overlaps among seeds. Comparing to OINS, our PRM-IMM algorithm runs slower, and this is mainly because the each MR-RR set for NIOS is much larger than the PW-RR set for OINS, but the number of RR sets needed in each case is not significantly different.

7.2.5 Alternative Objectives. Finally, we evaluate the two alternative objectives discussed in Section 6.4. Since RandomSeed&Round is clearly worse than other algorithms, while OneShot-IMM cannot assign seeds to multiple rounds, we do not include them as baselines in this part. We apply the idea in Section 6.4 to conduct these experiments for the OINS setting. Fig.7 shows the results on four datasets, and the results on the other four datasets are similar. Fig.7(a) shows the result for the Seed Minimization problem, in which we fix $T = 20$, and then use different algorithms to find the minimum budget that can make the popular ratio exceed 1. Fig.7(b) shows the result for the Round Minimization problem. For this problem, we need to set different budgets for different datasets based on our result from Fig.7(a) to balance the influence of T and k . The results in both figures consistently show that our algorithm can achieve the best surpass budget and surpass time compared to all other baselines.

8 FUTURE WORK

One interesting future direction of this work is to study adaptive strategies, where the seed selection of later rounds could depend on the actual promotional results of the earlier rounds. Another direction is to consider the delayed effect of promotion, where social network promotion may take multiple rounds to generate popularity growth. One may also look into the model where the promotion of the popular item and the novice item also competes in the social network.

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